

Linear Algebra I Recitation Class: Week 2

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denotes the sets of rational, real, and complex numbers, respectively. V is a vector space over a field F . If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind.

Recall that a subset W of V is a vector subspace of V if it satisfies the following conditions:

S1. $0_V \in W$.

S2. $w_1 + w_2 \in W$ for all $w_1, w_2 \in W$.

S3. $a \cdot w \in W$ for all $w \in W$, $a \in F$.

Here $+$, $\cdot$ are inherited from V .

Example (Base field matters!). Recall that the set $M_2(\mathbb{C})$ of 2×2 matrices in $\mathbb{C}$ is a vector space over $\mathbb{C}$, and also a vector space over $\mathbb{R}$. Define a subset of $M_2(\mathbb{C})$

$$\mathfrak{u}(2) := \{A \in M_2(\mathbb{C}) \mid A + A^* = 0\}$$

Here A^* is the **conjugate transpose** of A ,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^* := \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}, a, b, c, d \in \mathbb{C} \quad \overline{x + iy} := x - iy \text{ for any } x, y \in \mathbb{R}$$

After some computation, we deduce that

$$\mathfrak{u}(2) = \left\{ \begin{pmatrix} ai & p + qi \\ -p + qi & bi \end{pmatrix} \in M_2(\mathbb{C}) \mid a, b, p, q \in \mathbb{R} \right\}.$$

One can check that $\mathfrak{u}(2)$ is a real subspace of $M_2(\mathbb{C})$. However, it is NOT a complex vector subspace of $M_2(\mathbb{C})$. For example,

$$A = \begin{pmatrix} i & 0 \\ 0 & 0 \end{pmatrix} \in \mathfrak{u}(2), \text{ but } iA = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \notin \mathfrak{u}(2).$$

The subspace axiom S3 fails when the base field is $\mathbb{C}$.

We introduce several ways to obtain new vector (sub)spaces from vector (sub)spaces. From now on W, U will be vector subspaces of V unless we mention otherwise.

Exercise A. Show that $W \cap U$ is a vector subspace of V .

Proposition 1. [*FIS03*, Exercise 1.3.19] $W \cup U$ is a vector subspace of V if and only if $W \subseteq U$ or $U \subseteq W$.

Proof. ($\Leftarrow$) If $W \subseteq U$, then $W \cup U = U$, which is a vector subspace of V . Similarly, if $U \subseteq W$, then $W \cup U = W$, which is a vector subspace of V .

($\Rightarrow$) Suppose that $W \cup U$ is a vector subspace of V . Suppose on the contrary that neither $W \subseteq U$ nor $U \subseteq W$. Then there exist

$$w \in W \setminus U \quad \text{and} \quad u \in U \setminus W.$$

Since $u, w \in W \cup U$ and $W \cup U$ is a subspace, we have $u + w \in W \cup U$. If $u + w \in W$, then

$$u = (u + w) - w \in W,$$

contradicting $u \notin W$. If $u + w \in U$, then

$$w = (u + w) - u \in U,$$

contradicting $w \notin U$. Therefore, either $W \subseteq U$ or $U \subseteq W$. □

Definition 2. The **sum** of W, U is defined as

$$W + U := \{w + u \mid w \in W, u \in U\}.$$

In addition, if $W \cap U = \{0\}$, then we call $W + U$ a **direct sum**, denoted by $W \oplus U$.

Exercise B. Show that $W + U$ is a vector subspace of V . Furthermore, show that it is the smallest subspace containing both W and U , i.e. every subspace containing both W and U must contain $W + U$.

Proposition 3. [FIS03, Exercise 1.3.30] Let W and U be subspaces of a vector space V . Prove that V is the direct sum of W and U if and only if each vector in V can be uniquely written as $w + u$, where $w \in W$ and $u \in U$.

Proof. ($\Rightarrow$) Assume $V = W \oplus U$, then by the definition of sums of vector spaces, given $v \in V$, $v = w + u$ for some $w \in W, u \in U$ (this proves existence). Suppose $v = w + u = w' + u'$ for some $w, w' \in W, u, u' \in U$, then

$$w - w' = u' - u \in W \cap U = \{0\}.$$

That is, $w = w', u = u'$ (this proves uniqueness). □

Exercise C. Finish the proof.

Although the direct sum defined above is for two vector subspaces of the same vector space, we can also do the same things for a family of vector spaces.

Definition 4. Let I be a set and $\{V_i\}_{i \in I}$ be a family of vector spaces over F indexed by I .

1. The **direct product** of $\{V_i\}_{i \in I}$ is a set

$$\prod_{i \in I} V_i := \{v = (v_i)_{i \in I} \mid v_i \in V_i\}$$

with addition and scalar multiplication defined coordinatewise, i.e.

$$(v + w)_i := v_i + w_i \quad (a \cdot v)_i := a \cdot v_i$$

for any $i \in I, v = (v_i), w = (w_i) \in \prod_i V_i, a \in F$.

2. The **direct sum** of $\{V_i\}_{i \in I}$ is a subset of direct product defined by

$$\bigoplus_{i \in I} V_i := \left\{ (v_i)_{i \in I} \in \prod_{i \in I} V_i \mid v_i = 0 \text{ for all but finitely many } i \in I \right\}$$

where the addition and scalar multiplication are defined in the same way.

Remark. When the set I is finite, the direct sum and direct product coincides.

Exercise D. Prove that the direct product of $\prod_{i \in I} V_i$ is a vector space over F , and that the direct sum $\bigoplus_{i \in I} V_i$ is a vector subspace of $\prod_{i \in I} V_i$, hence a vector space over F .

Let $V = \mathbb{R}^2$ and let

$$W = \{(0, y) \mid y \in \mathbb{R}\}.$$

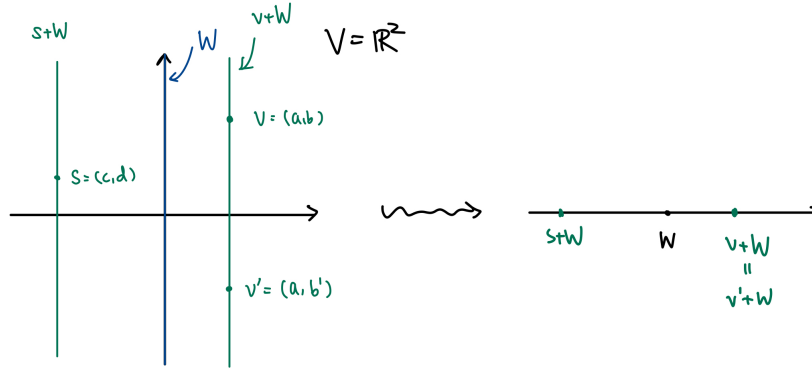
Then W is a vector subspace of V . Given $v = (a, b) \in V$, define

$$v + W := \{v + w \mid w \in W\}.$$

Geometrically, $v + W$ is the vertical line passing through v . If $v' = (a', b') \in V$, then

$$v + W = v' + W$$

if and only if $a = a'$. Thus a coset $v + W$ remembers only the x -coordinate of v . Consequently, the set of all cosets can be identified with the real line $\mathbb{R}$, so intuitively,



we can define addition and scalar multiplication on the set of $\{v + W \mid v \in V\}$ by

$$(u + W) + (v + W) := (u + v) + W \text{ and } a \cdot (v + W) := av + W \text{ for all } a \in \mathbb{R}. \quad (\star)$$

In fact, the operations are well-defined and together with the set, it forms a vector space. This is an example of quotient spaces.

Proposition 5. [FIS03, Exercise 1.3.31] Let W be a subspace of a vector space V . For any $v \in V$ the set $v + W = \{v + w \mid w \in W\}$ is called the **coset of W containing v** . Then

1. $v + W$ is a subspace of V if and only if $v \in W$.

2. $v_1 + W = v_2 + W$ if and only if $v_1 - v_2 \in W$
3. The operations defined in (★) are well-defined. That is, if $v_1 + W = v'_1 + W$ and $v_2 + W = v'_2 + W$ then

$$(v_1 + W) + (v_2 + W) = (v'_1 + W) + (v'_2 + W)$$

and

$$a(v_1 + W) = a(v'_1 + W)$$

for all $a \in F$.

4. $V/W := \{v + W \mid v \in V\}$ is a vector space over F , which is called the **quotient space of V modulo W** .

Proof. 1. ($\implies$) Suppose $v + W$ is a vector subspace of V , then $0_V \in v + W$, so $0_V = v + w$ for some $w \in W$. Hence $v = -w \in W$.

($\impliedby$) If $v \in W$, then $v + W = W$ is a subspace of V .

2. ($\implies$) Suppose $v_1 + W = v_2 + W$, then given $a \in v_1 + W$, say $a = v_1 + w$ for some $w \in W$. On the other hand, $a \in v_2 + W$, so $a = v_2 + w'$ for some $w' \in W$. Combining the equation and by cancellation law, we have $v_1 - v_2 = w' - w \in W$.

($\impliedby$) Suppose that $v_1 - v_2 \in W$. Write $v_1 - v_2 = w$ for some $w \in W$. Then

$$v_1 + W = (v_2 + w) + W = v_2 + (w + W) = v_2 + W,$$

since $w + W = W$.

3. If $v_1 + W = v'_1 + W$ and $v_2 + W = v'_2 + W$, then $v'_1 = v_1 + w_1$ and $v'_2 = v_2 + w_2$ for some $w_1, w_2 \in W$, and we have

$$\begin{aligned} (v'_1 + W) + (v'_2 + W) &= (v'_1 + v'_2) + W \\ &= (v_1 + w_1 + v_2 + w_2) + W \\ &= (v_1 + v_2) + (w_1 + w_2) + W \\ &= (v_1 + v_2) + W = (v_1 + W) + (v_2 + W). \end{aligned}$$

Similarly,

$$\begin{aligned} a(v'_1 + W) &= av'_1 + W \\ &= a(v_1 + w_1) + W \\ &= av_1 + aw_1 + W \\ &= av_1 + W \\ &= a(v_1 + W). \end{aligned}$$

4. Exercise.

□

If we only treat V/W as merely a set, then it is a special case of **quotient sets**¹.

Definition 6. Let X be a set, a **relation** R on X is a subset of $X \times X$. We say a is **R -related to b** when $(a, b) \in R$. In this case, we will also write aRb .

Example. Let $X = \{1, 2, 3, 4, 5\}$, if we define $xRy : \iff x \mid y$, then R can be represented as a subset

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 4), (3, 3), (4, 4), (5, 5)\} \subseteq X \times X$$

Definition 7. A relation $\sim$ on X is called an **equivalence relation** if it satisfies

1. (Reflexivity) $a \sim a$ for all $a \in X$.
2. (Symmetry) For all $a, b \in X$, if $a \sim b$, then $b \sim a$.
3. (Transitivity) For all $a, b, c \in X$, if $a \sim b$, and $b \sim c$, then $a \sim c$.

Example. 1. Let $X = \mathbb{R}$, then the relation $x \sim_e y : \iff x = y$ is an equivalence relation.

2. Let $X = G = (V, E)$ be an undirected graph, then the relation

$$v_i \sim_p v_j : \iff \text{there is a path from } v_i \text{ to } v_j$$

on V (we only care about the vertices) is an equivalence relation. Here we allow a path of length 0, so every vertex is connected to itself.

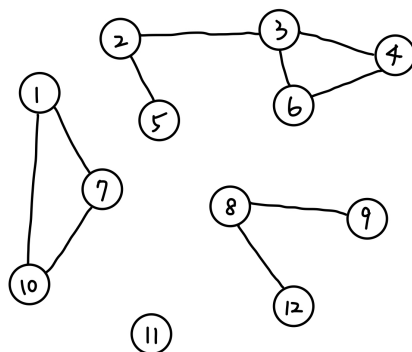


Figure 1: A graph

3. Let $X = \mathbb{Z}$, and fix a positive integer n , the relation

$$x \sim_n y : \iff x \equiv y \pmod{n} \iff n \mid x - y$$

is an equivalence relation.

¹They are also called quotient spaces but I do not want to make it more confusing...

4. Let $X = M_2(\mathbb{C})$ be the set of 2×2 matrices with coefficients in $\mathbb{C}$, then the relation

$$A \sim_s B : \iff A = PBP^{-1} \text{ for some } P \in \text{GL}_2(\mathbb{C})$$

is an equivalence relation. In this case, we say that A and B are **similar**². (Recall that $\text{GL}_2(\mathbb{C})$ is the set of 2×2 invertible matrices with coefficients in $\mathbb{C}$.)

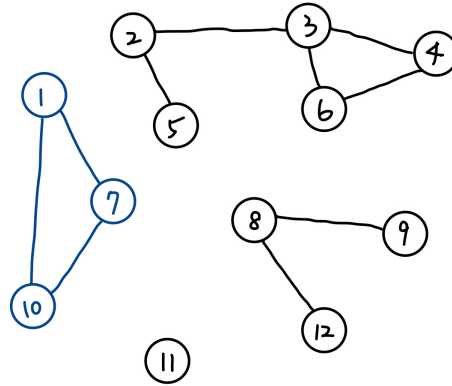
Exercise E. Let $X = V$ be a vector space over F , and W be a vector subspace of V , show that the relation $v \sim_W v' : \iff v - v' \in W$ is an equivalence relation.

Definition 8. Let X be a set, and $\sim$ be an equivalence relation on X , for any $x \in X$, the **equivalence class of x** is the set

$$[x]_{\sim} := \{y \in X \mid x \sim y\}.$$

Example. Using the notation of the preceding example:

1. The equivalence class of x in $\mathbb{R}$ is $\{x\}$.
2. The equivalence class of v_1 in V is the set of vertices in the subgraph in blue, which is called the **connected component** of v_1 .



3. The equivalence class of x in $\mathbb{Z}$ is called the **congruence class modulo n** .
4. The equivalence class of A in $M_2(\mathbb{C})$ is also called the **similarity class of A** .

Exercise F. Using the notation of the preceding exercise, show that the equivalence class of v is exactly $v + W$.

Definition 9. Let X be a set and $\sim$ be an equivalence relation on X , the **quotient set of X by $\sim$** is a set of subsets of X

$$X / \sim := \{[x]_{\sim} \mid x \in X\}$$

Remark. It is sometimes useful to treat each equivalence class as an element of $X / \sim$, or as a subset of X , depends on the situation.

²or **conjugate**, depending on the context on the context.

Example. Using the notation of the preceding example:

1. $\mathbb{R} / \sim_e = \{\{x\} \mid x \in \mathbb{R}\}$
2. $V / \sim_c$ is the set collecting all sets of vertices in each connected component.
3. $\mathbb{Z} / \sim_n = \{[0]_{\sim_n}, [1]_{\sim_n}, \dots, [n-1]_{\sim_n}\}$
4. $M_2(\mathbb{C}) / \sim_s = \left\{ \left[\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right]_{\sim_s}, \left[\begin{pmatrix} c & 1 \\ 0 & c \end{pmatrix} \right]_{\sim_s} \mid a, b, c \in \mathbb{C} \right\}$. Note that this list is not a unique parametrization, because

$$\left[\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right]_{\sim_s} = \left[\begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix} \right]_{\sim_s}$$

This is a special case of the **Jordan canonical form**, which you will study near the end of the linear algebra course. Exciting, isn't it?

Exercise G. Using the notation of the preceding exercise, show that the quotient set $V / \sim_W$ is exactly V/W . (This should be immediate once you are familiar with the definitions.)

One might observe from the graph that all set of vertices of each connected component are pairwise disjoint, and their union is the set of all vertices. This also holds true for any quotient set of a set by an equivalence relation.

Proposition 10. *Let X be a set and $\sim$ be an equivalence relation on X , then*

1. *For any $x, y \in X$, either $[x]_{\sim} \cap [y]_{\sim} = \emptyset$ or $[x]_{\sim} = [y]_{\sim}$ occurs.*
2. $X = \bigcup_{x \in X} [x]_{\sim}$.

Here we see equivalence classes as subsets of X . In this case, we say that the set of all equivalence classes of X forms a **partition** of X .

Proof. Let $x, y \in X$, if $[x]_{\sim} \cap [y]_{\sim} = \emptyset$, then the proof is done. Otherwise, there exists an element $z \in [x]_{\sim} \cap [y]_{\sim}$. That is, $z \sim x$ and $z \sim y$, by the transitivity and symmetry of equivalence relation, we have $x \sim y$. Thus, for any $\alpha \in [x]_{\sim}$, $\alpha \sim x$ so $\alpha \sim y$ (this proves $[x]_{\sim} \subseteq [y]_{\sim}$). Similarly, for any $\beta \in [y]_{\sim}$, $\beta \sim y$, so $\beta \sim x$ (this proves $[x]_{\sim} \supseteq [y]_{\sim}$). We have shown the first statement. The second one is clear since any union of subsets of X is a subset of X , and each element lies in some equivalence class (e.g. the equivalence class of itself). $\square$

The proof of the first statement can also be used to prove the following facts:

Proposition 11. *Let X be a set and $\sim$ be an equivalence relation on X . TFAE³:*

1. $x \sim y$

³Abbreviation of "The following are equivalent"

$$2. [x]_{\sim} \cap [y]_{\sim} \neq \emptyset$$

$$3. [x]_{\sim} = [y]_{\sim}$$

Exercise H. Prove the above proposition. (Hint: Prove 2 implies 1, 1 implies 3, and 3 implies 2.)

Corollary 12. *Let V be a vector space over F and W be a vector subspace of V , then V/W (regarded as a set of subsets of V) forms a partition of V .*

Proof. This is immediate from Proposition 10. □

References

- [FIS03] Stephen H. Friedberg, Arnold J. Insel, and Lawrence E. Spence. *Linear algebra*. Prentice Hall, Inc., Upper Saddle River, NJ, fourth edition, 2003.
- [Fra02] John B. Fraleigh. *A first course in abstract algebra*. Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 2002.
- [Hun80] Thomas W. Hungerford. *Algebra*, volume 73 of *Graduate Texts in Mathematics*. Springer-Verlag, New York-Berlin, 1980. Reprint of the 1974 original.